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Big Data and Data Analytics**

Entitled

**Big Data Analytics and Sentiment Mining on Amazon Movies &
TV Reviews: Uncovering Consumer Patterns**

By

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Abstract:

This project focuses on the data analysis of customer reviews from the Amazon Movies & TV category, with the objective of extracting meaningful insights from large-scale textual data. The main challenge lies in processing and analyzing a massive volume of unstructured reviews to evaluate customer satisfaction and support decision-making.

To address this problem, a Big Data approach was adopted, combining data engineering, natural language processing (NLP), and machine learning techniques. The dataset was cleaned and transformed, and text data was converted into numerical representations using TF-IDF. Classification models were then applied to perform sentiment analysis and categorize reviews as positive, negative, or neutral.

The results highlight key trends in customer behavior, including sentiment distribution, review patterns over time, and product performance. An interactive dashboard was developed using Power BI to visualize these insights effectively.

This project demonstrates how data-driven approaches can enhance understanding of customer feedback and support strategic business decisions.

Keywords: Data Analysis, Customer Reviews, Sentiment Analysis, Big Data, Machine Learning, Amazon Reviews, Data Visualization

Résumé:

Ce projet porte sur l'analyse des données des avis clients issus de la catégorie Amazon Movies & TV, avec pour objectif d'extraire des informations pertinentes à partir de données textuelles massives. Le principal défi réside dans le traitement et l'analyse d'un grand volume de données non structurées afin d'évaluer la satisfaction des clients et d'aider à la prise de décision.

Pour répondre à cette problématique, une approche Big Data a été adoptée, combinant l'ingénierie des données, le traitement du langage naturel (NLP) et des techniques de machine learning. Les données ont été nettoyées et transformées, puis les textes ont été convertis en représentations numériques à l'aide de la méthode TF-IDF. Des modèles de classification ont ensuite été utilisés pour réaliser une analyse de sentiment et catégoriser les avis en positifs, négatifs ou neutres.

Les résultats mettent en évidence des tendances clés du comportement des clients, notamment la distribution des sentiments, l'évolution des avis dans le temps et la performance des produits. Un tableau de bord interactif a été développé en utilisant Power BI afin de visualiser efficacement ces informations.

Ce projet démontre l'importance des approches basées sur les données pour mieux comprendre les retours clients et soutenir la prise de décision stratégique.

Mots-clés: Analyse de données, Avis clients, Analyse de sentiment, Big Data, Machine Learning, Avis Amazon, Visualisation de données

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General Introduction

The rapid growth of digital platforms and e-commerce services has led to the generation of massive amounts of customer-generated data. Among the most valuable forms of this data are customer reviews, which reflect users' opinions, experiences, and satisfaction levels regarding products and services.

Analyzing this information has become increasingly important for organizations seeking to improve customer satisfaction, optimize decision-making processes, and better understand consumer behavior.

However, the large volume and unstructured nature of textual reviews make manual analysis difficult and inefficient. This challenge has encouraged the adoption of Big Data technologies, Natural Language Processing (NLP), and Machine Learning techniques capable of automatically processing and analyzing customer feedback at scale.

The objective of this project is to develop a Big Data analytics system for Amazon Movies & TV customer reviews in order to automatically evaluate customer sentiment and generate meaningful business insights. The project combines data engineering, NLP preprocessing, machine learning classification, and interactive data visualization techniques.

The implemented system processes large-scale review data through multiple stages, including data cleaning, sentiment label generation, text preprocessing, TF-IDF feature engineering, and machine learning-based sentiment classification. In addition, interactive dashboards were developed using Power BI to visualize review trends, customer behavior, sentiment distribution, and product performance indicators.

This project demonstrates how Big Data technologies and machine learning techniques can transform unstructured textual information into actionable insights that support data-driven decision-making processes.

The report is organized into four main chapters:

- Chapter 1 presents the project context and the state of the art related to Big Data, sentiment analysis, and business intelligence.
- Chapter 2 describes the Big Data architecture, data engineering pipeline, preprocessing operations, and NLP techniques used throughout the project.
- Chapter 3 focuses on the machine learning models implemented for sentiment classification and their evaluation results.
- Chapter 4 presents the business intelligence dashboards, visualizations, and analytical insights generated using Power BI.

Chapter 1:

Project Context & State Of The Art

1/ Introduction

In recent years, the rapid growth of e-commerce platforms has led to an exponential increase in user-generated content, particularly in the form of customer reviews. These reviews represent a valuable source of information, reflecting customers' opinions, experiences, and satisfaction levels regarding products and services. Among these platforms, Amazon has become one of the largest sources of online reviews, covering a wide range of product categories, including Movies and TV shows.

However, the massive volume and unstructured nature of these textual data make manual analysis extremely difficult and time-consuming. As a result, organizations face significant challenges in extracting meaningful insights that can support strategic decision-making and improve customer experience.

In this context, data analysis techniques combined with Big Data technologies and machine learning methods provide effective solutions for processing and analyzing large-scale datasets. In particular, Natural Language Processing (NLP) enables the transformation of textual data into structured information that can be analyzed and interpreted.

This project focuses on the analysis of customer reviews from the Amazon Movies & TV category. The main objective is to extract valuable insights from large volumes of textual data, evaluate customer sentiment, and identify patterns in user behavior. To achieve this, a complete data pipeline is implemented, including data preprocessing, feature engineering, sentiment analysis, and interactive data visualization.

The results of this project aim to provide a better understanding of customer feedback and demonstrate how data-driven approaches can support informed decision-making in a business context.

2/ Problem Statement

The rapid growth of e-commerce and digital streaming platforms has led to the generation of massive amounts of customer feedback in the form of online reviews. Amazon Movies & TV products receive thousands of reviews from customers expressing their opinions, experiences, and satisfaction levels regarding various products and services.

These reviews represent a highly valuable source of information for understanding customer behavior and evaluating product quality. However, due to the large volume and unstructured nature of textual reviews, extracting meaningful insights manually becomes extremely difficult and time-consuming.

Traditional manual analysis methods are inefficient when dealing with Big Data environments containing thousands or millions of customer reviews. In addition, customer opinions are often expressed using informal language, mixed sentiments, abbreviations, and context-dependent expressions, which further complicates analysis tasks.

Organizations require efficient analytical systems capable of automatically processing large-scale textual data, identifying customer sentiment, and generating actionable insights that support decision-making processes.

Therefore, there is a growing need for intelligent systems capable of combining Big Data technologies, Natural Language Processing, and Machine Learning techniques in order to automate customer review analysis and improve business intelligence capabilities.

3/ Project Objectives

The primary objective of this project is to develop a system capable of analyzing customer reviews at scale and extracting valuable insights that can support decision-making.

The specific objectives are as follows:

- To collect and manage a large dataset of customer reviews from the Amazon Movies & TV category
- To preprocess and clean textual data in order to prepare it for analysis
- To apply Natural Language Processing techniques to transform text into structured data
- To implement machine learning models for sentiment analysis
- To evaluate the performance of the models using appropriate metrics
- To design and develop an interactive dashboard for data visualization and analysis

4/Literature Review

4.1/ Text Mining and Natural Language Processing(NLP)

Text mining refers to the process of extracting meaningful information from textual data. It involves several steps, including tokenization, stopword removal, and text normalization. Natural Language Processing (NLP) plays a crucial role in enabling machines to understand and process human language.

NLP techniques allow the transformation of unstructured text into structured formats that can be analyzed using computational methods. These techniques are widely used in applications such as sentiment analysis, information retrieval, and text classification.

4.2/ Sentiment Analysis Techniques

Sentiment analysis is a subfield of NLP that focuses on identifying and extracting opinions expressed in text. It aims to determine whether a given piece of text is positive, negative, or neutral. There are several approaches to sentiment analysis:

- Rule-based approaches, which rely on predefined dictionaries and linguistic rules
- Machine learning approaches, which use labeled data to train models capable of classifying text automatically

Machine learning methods are generally more flexible and accurate, especially when dealing with large datasets.

4.3/ Machine Learning Approaches

Machine learning plays a key role in text classification tasks. Algorithms such as Logistic Regression and Linear SVC are commonly used for sentiment analysis due to their efficiency and simplicity.

These models require numerical input, which means that textual data must first be converted into numerical features. One of the most widely used techniques for this purpose is TF-IDF (Term Frequency–Inverse Document Frequency), which measures the importance of words in a document relative to a collection of documents.

5/ Limitations Of Existing Solutions

Several traditional approaches have been used to analyze customer reviews and evaluate customer satisfaction. However, many existing solutions present important limitations when dealing with large-scale textual data.

Manual analysis methods require significant human effort and become impractical when processing massive datasets containing thousands of reviews. These approaches are also time-consuming and may produce inconsistent interpretations depending on the analyst.

Conventional Business Intelligence tools are generally designed for structured numerical data and often struggle to process unstructured textual information effectively. Without advanced Natural Language Processing techniques, extracting meaningful sentiment information from raw customer reviews remains difficult.

In addition, simple keyword-based approaches frequently fail to capture contextual meaning within customer opinions. Words may express different sentiments depending on context, especially in the presence of negation, sarcasm, or mixed emotional expressions.

Many traditional systems also face scalability limitations when handling Big Data environments. Processing large volumes of customer reviews requires efficient storage, distributed processing capabilities, and optimized analytical workflows.

Furthermore, certain existing sentiment analysis approaches rely heavily on manually defined rules, which reduces flexibility and limits their ability to generalize across different datasets and review contexts.

These limitations highlight the need for more scalable, intelligent, and automated analytical systems capable of efficiently processing large-scale customer review data.

6/ Proposed Solution

To address the limitations of existing approaches, this project proposes a Big Data analytics and sentiment analysis system designed to automatically process and analyze Amazon Movies & TV customer reviews.

The proposed solution combines multiple technologies and analytical techniques, including Big Data processing, Natural Language Processing (NLP), Machine Learning, and Business Intelligence visualization tools.

The system begins by collecting and organizing customer review data and metadata information. Data preprocessing operations are then applied in order to clean and standardize the textual content. Several NLP techniques are used, including stopword removal, text normalization, tokenization, and negation handling, in order to improve textual data quality.

After preprocessing, TF-IDF feature engineering is applied to transform customer reviews into numerical representations suitable for machine learning models.

Two supervised machine learning algorithms, Logistic Regression and Linear Support Vector Classifier (Linear SVC), are implemented to automatically classify customer reviews into positive, neutral, and negative sentiments.

The project also integrates PySpark-based distributed processing in order to demonstrate scalable Big Data analytical capabilities for handling large-scale review datasets efficiently.

Finally, Power BI dashboards are developed to visualize customer review trends, sentiment distributions, product performance indicators, and business insights through interactive charts and KPIs.

The proposed solution aims to automate customer feedback analysis, improve analytical efficiency, and support data-driven decision-making processes through intelligent and scalable technologies.

7/ Chapter Conclusion

This chapter introduced the general context of the project and highlighted the importance of analyzing customer reviews in the era of Big Data. The problem statement and objectives were clearly defined, and an overview of existing techniques in text mining, sentiment analysis, and machine learning was presented. Finally, the limitations of current solutions were discussed, leading to the presentation of the proposed approach.

The next chapter will focus on the Big Data architecture and data engineering processes used to prepare and manage the dataset.

Chapter 2:

Big Data Architecture & Data Engineering

1/ Introduction

This chapter presents the architecture and data engineering processes implemented throughout the development of this project. Since the dataset contains a large volume of unstructured textual data collected over multiple years, it was necessary to use several preprocessing, transformation, and analysis techniques in order to prepare the data for machine learning and business intelligence tasks. The main objective of this phase was to transform raw customer reviews into a clean, structured, and analyzable dataset capable of generating meaningful business insights. This involved handling missing and duplicated values, applying Natural Language Processing (NLP) techniques, extracting useful textual features, and preparing the data for sentiment classification.

In addition, Big Data technologies and scalable processing tools were used to efficiently manipulate and analyze large datasets. The integration of machine learning and data visualization techniques further enhanced the ability to extract valuable insights from customer feedback.

This chapter describes the dataset structure, the technologies used, the preprocessing pipeline, the NLP operations applied to customer reviews, and the feature engineering methods used to prepare the data for predictive modeling.

2/ Dataset Description

The dataset used in this project consists of customer reviews from the Amazon Movies & TV category. It contains approximately **1.7 million reviews**, making it a large-scale dataset suitable for Big Data analysis.

Each record in the dataset includes several attributes, the most important being:

- **reviewText**: the textual content of the review
- **rating**: a numerical score between 1 and 5 assigned by the customer
- **reviewTime**: the date when the review was posted
- **asin**: a unique identifier for each product
- **title**: the name of the product
- **sentiment**: a derived label indicating whether the review is positive, negative, or neutral
- **clean_text**: the processed version of the original review text

Initially, the reviews dataset did not always contain accurate or complete product titles, with many entries labeled as “Unknown Product.” To address this limitation, an additional dataset containing product metadata (referred to as the media dataset) was integrated into the project.

A data merging process was performed using the ASIN (Amazon Standard Identification Number) as a common key between the two datasets. This allowed the enrichment of the reviews dataset with correct and complete product names.

This data integration step significantly improved the quality of the dataset, enabling more accurate analysis, especially for product-level insights and visualizations.

Overall, the final dataset combines both structured and enriched information, providing a solid foundation for advanced data analysis and machine learning tasks.

It is important to note that the original dataset did not include a predefined sentiment label. In order to perform supervised machine learning, a new variable called sentiment was created based on the rating provided by users.

The sentiment was derived using a rule-based approach, where ratings above a certain threshold were classified as positive, lower ratings as negative, and intermediate values as neutral. This transformation allowed the creation of labeled data necessary for training classification models.

The generated sentiment labels serve as the ground truth for the machine learning phase. The objective of the models is to learn patterns from the textual content of the reviews and predict the corresponding sentiment. The performance of these models is then evaluated by comparing the predicted sentiment with the assigned labels.

_id	ObjectId	reviewText	String	rating	Int32	reviewTime	String	asin	String	title	String	sentiment	String	clean_text	String
1	ObjectId('69ac35c847d345...	"This is a charming vers...		4		"02 26, 2008"		"0005019281"		"An American Christmas C...		"positive"		"charming version classi...	
2	ObjectId('69ac35c847d345...	"It was good but not as ...		3		"12 30, 2013"		"0005019281"		"An American Christmas C...		"neutral"		"good emotionally moving...	
3	ObjectId('69ac35c847d345...	"Don't get me wrong, Win...		3		"12 30, 2013"		"0005019281"		"An American Christmas C...		"neutral"		"dont get wrong winkler ...	
4	ObjectId('69ac35c847d345...	"Henry Winkler is very g...		5		"02 13, 2008"		"0005019281"		"An American Christmas C...		"positive"		"henry winkler good tel...	
5	ObjectId('69ac35c847d345...	"This is one of the best...		4		"12 22, 2013"		"0005019281"		"An American Christmas C...		"positive"		"one best scrooge movie ...	

Figure 2.1: Sample of the Amazon Reviews Dataset

3/ **Big Data Architecture**

The architecture of the project was designed to support the efficient processing and analysis of large-scale textual data.

The workflow of the system consists of multiple interconnected stages that transform raw customer reviews into business insights and predictive models.

The architecture includes the following stages:

- 1.Data collection and loading
- 2.Data cleaning and preprocessing
- 3.Natural Language Processing (NLP)
- 4.Feature engineering using TF-IDF
- 5.Machine learning model training
- 6.Prediction and evaluation
- 7.Data visualization and business analysis using Power BI

The first stage involves loading and preparing the datasets for processing. Once loaded, the data undergoes several cleaning operations to improve quality and consistency.

After preprocessing, NLP techniques are applied to clean and normalize customer reviews. The textual data is then transformed into numerical representations using TF-IDF vectorization. The transformed data is used to train machine learning models capable of classifying customer sentiment. Finally, the processed data and analysis results are visualized through interactive dashboards developed using Power BI.

This architecture combines Big Data processing, NLP, machine learning, and business intelligence technologies into a unified analytical pipeline.

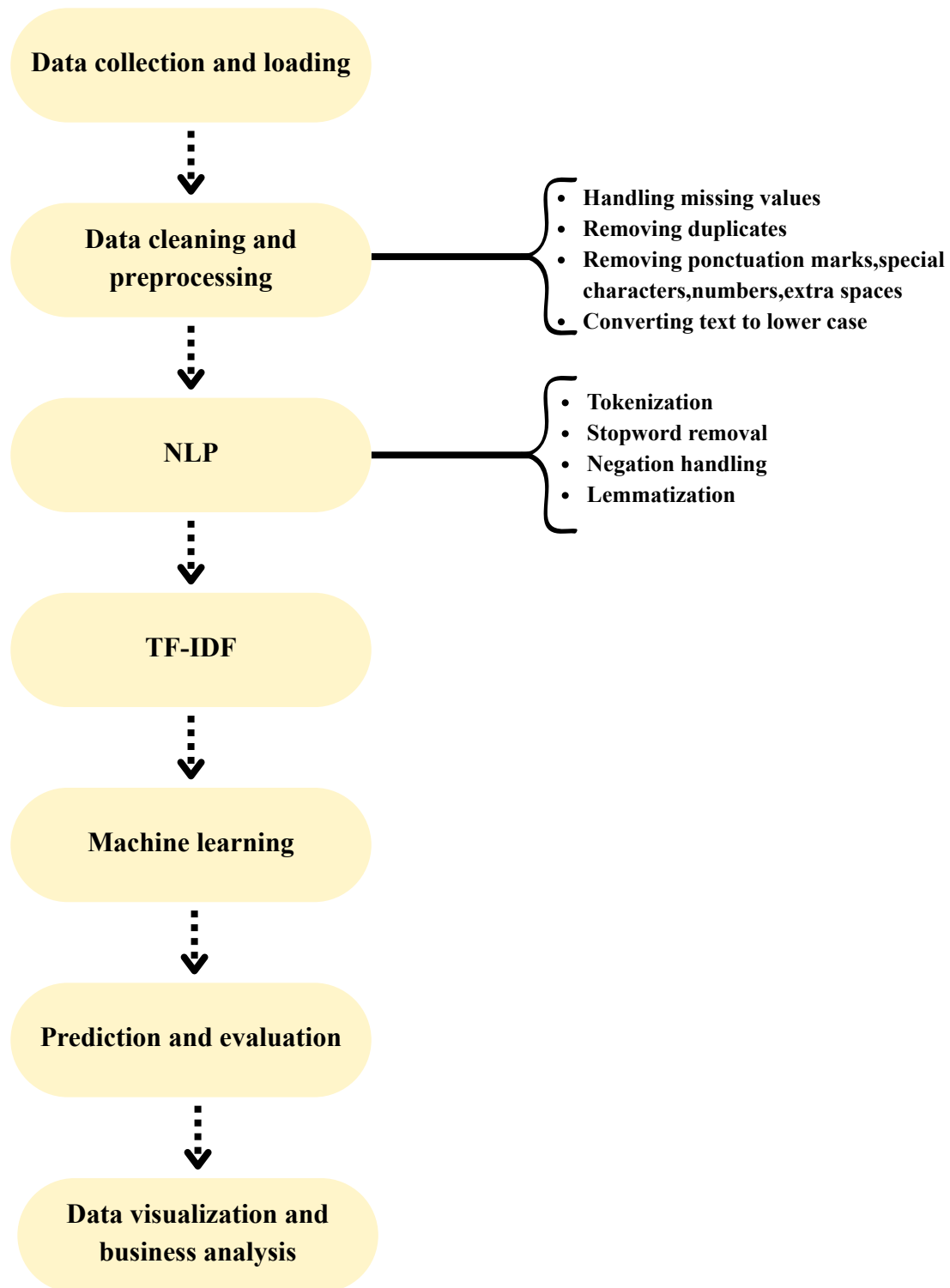


Figure 2.2: Project Architecture Pipeline

PySpark-Based Big Data Processing

Although the majority of preprocessing, NLP operations, and machine learning tasks were implemented using standard Python libraries, an additional Big Data processing phase was implemented using PySpark.

The objective of this phase was to demonstrate how large-scale customer review data can be processed using distributed computing techniques commonly used in Big Data environments. Using PySpark, several analytical operations and Key Performance Indicators (KPIs) were reproduced, including Total number of reviews, Average rating, Sentiment distribution, etc

This parallel implementation allowed a comparison between traditional data processing approaches and distributed Big Data processing techniques.

PySpark was particularly useful for handling large datasets efficiently through distributed computation and scalable data manipulation. While the complete machine learning pipeline was implemented using standard Python tools, the PySpark phase demonstrated the applicability of Big Data technologies for large-scale analytical tasks.

4/ Technologies and Tools Used

Several technologies and software tools were used throughout the project. Each tool played an important role in data preprocessing, analysis, modeling, or visualization.

Python

Python was used as the primary programming language for this project due to its flexibility and extensive ecosystem of data science libraries.

Python was used for:

- Data preprocessing
- NLP operations
- Machine learning implementation
- Feature engineering
- Data manipulation and analysis

Python's simplicity and strong library support make it highly suitable for Big Data and machine learning applications.

Pandas

The library was used for structured data manipulation and preprocessing tasks.

Pandas provides efficient tools for:

- DataFrame manipulation
- Data filtering
- Aggregation operations
- Statistical analysis
- Dataset merging

The library was particularly useful for exploratory data analysis and preparing smaller subsets of data for machine learning tasks.

Pymongo

PyMongo was used as the interface between Python and MongoDB for managing and interacting with large-scale review data. MongoDB, a NoSQL database system, is particularly suitable for handling semi-structured and unstructured data such as customer reviews.

Through PyMongo, the project was able to efficiently store, retrieve, and manipulate review records in a flexible document-oriented structure. Unlike traditional relational databases, MongoDB allows scalable storage of textual data without requiring a rigid schema, making it highly appropriate for Big Data applications.

PyMongo facilitated data access operations and simplified the integration of review datasets into the analytical workflow. The use of MongoDB technologies in this project also demonstrates the applicability of NoSQL database systems in Big Data environments where large volumes of heterogeneous data must be processed efficiently.

The integration of PyMongo contributed to the overall scalability and flexibility of the data engineering pipeline implemented in this project.

NLTK

The Natural Language Toolkit (NLTK) library was used to perform several Natural Language Processing (NLP) tasks during the preprocessing phase of the project. Since customer reviews are unstructured textual data, NLP techniques were essential for cleaning and preparing the text before machine learning analysis:

- tokenization
- stopwords removal
- lemmatization

NLTK was primarily used for stopwords removal and text processing operations. Common words that provide little analytical value, such as “the”, “is”, and “and”, were removed in order to reduce noise and improve the quality of textual features.

Special attention was given to preserving important negation words such as “not”, “no”, and “but”, since they play a major role in sentiment interpretation. This helped maintain the contextual meaning of customer opinions and improved the reliability of sentiment classification.

The use of NLTK significantly improved the preprocessing pipeline by producing cleaner and more meaningful textual data for feature extraction and machine learning tasks.

Scikit-learn

The library was used for machine learning implementation and model evaluation.

Scikit-learn provided the tools necessary for:

- TF-IDF vectorization
- Dataset splitting
- Logistic Regression implementation
- Linear SVC implementation
- Model evaluation and performance metrics

The library simplifies machine learning workflows and provides reliable implementations of popular algorithms.

PySpark

PySpark was used to process and manipulate large-scale datasets efficiently.

Since the Amazon reviews dataset contains a significant volume of customer reviews collected over many years, PySpark enabled scalable data processing and improved computational performance.

PySpark operations included:

Loading large datasets

- Data filtering and transformation
- Handling missing values
- Removing duplicates
- Aggregating review statistics
- Preparing structured outputs for analysis

One of the major advantages of PySpark is its distributed processing capability, which allows large datasets to be processed more efficiently compared to traditional approaches.

PySpark is widely used in Big Data environments because it supports parallel processing and can handle massive amounts of data efficiently.

Power BI

The business intelligence software was used to create interactive dashboards and visualizations.

Power BI enabled the transformation of analytical results into business-oriented insights through:

- Interactive dashboards
- KPI indicators
- Trend analysis charts
- Product performance visualizations
- Sentiment analysis reporting

The use of Power BI improved the interpretability of the project and allowed the results to be presented in a professional and user-friendly way.

5/ Data Cleaning and Preprocessing

Data cleaning and preprocessing represent critical stages in any data science and Big Data project. Raw datasets often contain incomplete, inconsistent, duplicated, or noisy information that can negatively affect analytical accuracy and machine learning performance.

Since the Amazon Movies & TV Reviews dataset contains a very large volume of customer-generated textual data collected over multiple years, several preprocessing operations were required before performing sentiment analysis and predictive modeling.

The main objective of this phase was to improve data quality, ensure consistency, and prepare the dataset for Natural Language Processing (NLP) and machine learning tasks. Several preprocessing techniques were applied throughout this stage.

5.1/ Handling Missing Values

The first preprocessing step involved identifying and handling missing values within the dataset. Missing data is a common issue in large-scale datasets and may lead to inaccurate analysis results if not properly addressed. Important columns such as review text, ratings, and product identifiers were analyzed to detect incomplete records.

Rows containing missing or null values in critical fields were removed to maintain data consistency and ensure the reliability of the analytical process.

Handling missing values is essential because machine learning models require complete and valid input data in order to produce accurate predictions.

If a review is empty it returns a void
The empty titles were replaced with “Unknown Product”.

5.2/ Removing Duplicates

Duplicate records were identified and removed from the dataset during preprocessing. In large-scale review platforms, duplicate entries may occur due to data collection processes, repeated submissions, or synchronization issues. Keeping duplicated reviews could distort statistical analysis and artificially influence sentiment distributions.

Removing duplicate records helps ensure that:

- Each customer review contributes only once to the analysis
- Statistical metrics remain accurate
- Machine learning models are trained on reliable data
- This step improved the overall quality and integrity of the dataset.

5.3/ Sentiment Label Creation

The original Amazon dataset did not contain explicit sentiment labels indicating whether reviews were positive, neutral, or negative.

To enable sentiment classification, a new sentiment column was created based on customer ratings. The following rules were applied:

Ratings $\geq 4 \rightarrow$ Positive

Rating = 3 $\rightarrow$ Neutral

Ratings $\leq 2 \rightarrow$ Negative

This generated sentiment label was later used as the target variable for machine learning classification models.

The creation of sentiment labels allowed the project to transform numerical ratings into categorical sentiment information suitable for predictive analysis.

However, it is important to note that numerical ratings do not always perfectly represent the true emotional sentiment expressed in review text. Some reviews may contain mixed or contradictory opinions despite their assigned rating.

5.4/ Text Cleaning

Customer reviews contain large amounts of noisy and inconsistent textual information that can negatively impact machine learning performance.

Several text cleaning operations were therefore applied to standardize the review content before performing NLP and feature extraction tasks.

The cleaning process included:

- Conversion of text to lowercase
- Removal of punctuation marks and numbers
- Removal of special characters
- Removal of unnecessary spaces

For example:

"Great Movie!!! Highly Recommended."

$\rightarrow$ "great movie highly recommended"

These transformations reduced textual variability and simplified the feature extraction process.

Text cleaning is particularly important in sentiment analysis because machine learning algorithms require consistent and structured textual input.

5.5/ Importance of Preprocessing

Data preprocessing plays a fundamental role in the success of machine learning and NLP applications.

Without proper preprocessing:

- Textual noise may reduce prediction accuracy
- Duplicate and missing records may distort analysis results
- Inconsistent formatting may affect feature extraction
- Irrelevant information may increase computational complexity

The preprocessing phase implemented in this project significantly improved dataset quality and created a reliable foundation for subsequent NLP operations, TF-IDF vectorization, and sentiment classification tasks.

By cleaning and structuring the data properly, the project achieved more accurate analytical results and improved machine learning performance.

6/ Natural Language Processing (NLP)

Since customer reviews are unstructured textual data, Natural Language Processing (NLP) techniques were applied to prepare the text for analysis and machine learning tasks. NLP plays a fundamental role in sentiment analysis because machine learning models cannot directly interpret raw human language efficiently.

Customer reviews often contain inconsistent formatting, punctuation, abbreviations, irrelevant words, and contextual expressions that may negatively affect model performance if not properly processed. Therefore, several NLP preprocessing operations were implemented to clean, normalize, and structure the textual data.

The objective of this phase was to reduce textual noise while preserving meaningful information that contributes to sentiment classification.

6.1/ Tokenization

Tokenization is the process of splitting textual content into smaller units called tokens, which are usually individual words.

This step allows textual data to be processed more efficiently by machine learning algorithms. Instead of treating an entire review as a single block of text, tokenization breaks the sentence into meaningful components that can later be analyzed individually.

For example:

"This movie was very good"

→ ["this", "movie", "was", "very", "good"]

Tokenization simplifies subsequent NLP operations such as stopwords removal and feature extraction.

In this project, tokenization was applied after text cleaning operations to ensure that reviews were processed in a consistent and structured format.

6.2/ Stopword Removal

Stopwords are very common words that appear frequently in textual data but usually provide limited analytical value. Examples include:

- the
- is
- and
- of
- to

Removing stopwords helps reduce unnecessary noise in the dataset and improves the efficiency of machine learning models.

The NLTK library was used to identify and remove English stopwords from customer reviews. This preprocessing step reduced the dimensionality of the textual data and allowed the models to focus more on informative and sentiment-related words.

For example:

"This movie is very good and enjoyable"

→ **"movie good enjoyable"**

Stopword removal contributes to improving text quality and reducing computational complexity during feature extraction.

6.3/ Negation Handling

One of the major challenges in sentiment analysis is handling negation words such as:

- not
- no
- never
- but

These words significantly influence the meaning and sentiment of a sentence.

For example:

"good" → positive sentiment

"not good" → negative sentiment

If negation words are removed during preprocessing, the contextual meaning of customer opinions may be distorted, leading to incorrect sentiment predictions.

To address this issue, important negation terms were intentionally preserved during the stopwords removal process. This allowed the sentiment analysis models to better capture contextual meaning and improve classification reliability.

Handling negation correctly is essential in NLP because customer opinions often rely heavily on contextual expressions and linguistic nuances.

6.4/ Lemmatization

Lemmatization is an important Natural Language Processing (NLP) technique used to reduce words to their base or dictionary form, known as the lemma. Unlike stemming, which simply removes prefixes or suffixes, lemmatization considers the grammatical meaning and context of a word before transforming it. This process helps preserve the actual meaning of words while reducing vocabulary complexity in textual data.

For example:

- running → run
- **better** → **good**
- **studies** → **study**

In text mining and sentiment analysis tasks, multiple forms of the same word may appear frequently in the dataset. Without lemmatization, these variations are treated as separate terms, which increases dimensionality and may reduce model performance. By converting words into their root forms, lemmatization improves text consistency and helps machine learning algorithms better understand relationships between words.

Lemmatization was applied during the preprocessing stage after tokenization and stopword removal. This step ensured that semantically similar words were represented uniformly throughout the dataset. As a result, the textual data became cleaner, more structured, and more suitable for feature extraction techniques such as TF-IDF.

6.5/ NLP Challenges

Despite the preprocessing techniques applied, Natural Language Processing remains a complex task due to the ambiguity and variability of human language.

Several challenges were encountered during the project:

- Context-dependent expressions
- Sarcasm and irony
- Negation handling
- Informal language and abbreviations
- Reviews containing mixed sentiments
- Ratings that do not perfectly match textual sentiment

For example, a review may contain positive words while expressing an overall negative opinion: "It was good but not emotionally engaging."

In such cases, contextual interpretation becomes difficult for traditional machine learning models. Additionally, sentiment labels were generated from numerical ratings rather than manually annotated textual sentiment, which may introduce inconsistencies between written opinions and assigned labels.

These challenges highlight the limitations of traditional NLP approaches and demonstrate the complexity of sentiment analysis tasks involving large-scale customer review data.

Despite these limitations, the NLP preprocessing pipeline significantly improved text quality and contributed to better machine learning performance.

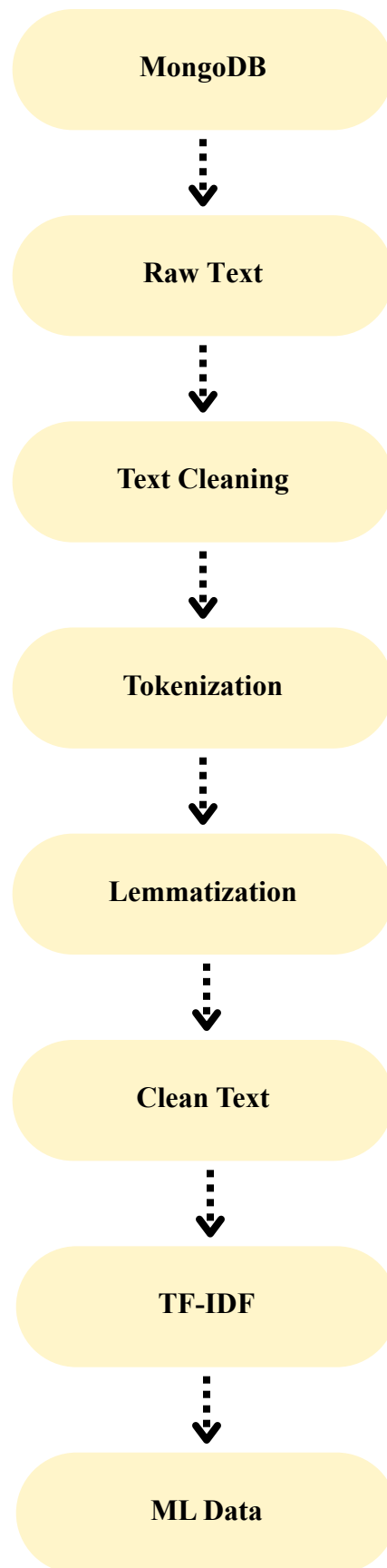


Figure 2.3: NLP system pipeline

7/ Feature Engineering

After preprocessing and NLP operations, the customer reviews still consisted of textual data that could not be directly interpreted by machine learning algorithms. Machine learning models require numerical input rather than raw text. Therefore, feature engineering techniques were applied to transform textual reviews into structured numerical representations.

Feature engineering is one of the most important stages in Natural Language Processing and machine learning projects because the quality of extracted features directly affects model performance and prediction accuracy.

In this project, the TF-IDF (Term Frequency–Inverse Document Frequency) technique was used to convert customer reviews into numerical feature vectors suitable for sentiment classification.

7.1/TF-IDF Vectorization

TF-IDF stands for Term Frequency–Inverse Document Frequency. It is a statistical technique used to measure the importance of words within a document relative to an entire collection of documents.

The main objective of TF-IDF is to identify words that carry meaningful information while reducing the influence of overly common words.

TF-IDF is composed of two main components:

- **Term Frequency (TF)**

Term Frequency measures how often a word appears within a specific document or review. Words that appear frequently within a review receive higher importance scores because they are likely related to the main content or sentiment expressed by the customer.

For example, if the word “excellent” appears multiple times in a review, it may indicate strong positive sentiment.

- **Inverse Document Frequency (IDF)**

Inverse Document Frequency reduces the importance of words that appear too frequently across all reviews.

Words that occur in nearly every document provide limited analytical value because they do not help distinguish between sentiments or topics.

For example, common words such as:

- movie
- film
- time

may appear in many reviews and therefore receive lower importance scores.

The combination of TF and IDF allows the algorithm to emphasize informative words while minimizing irrelevant or overly common terms.

The TF-IDF vectorization process transforms each review into a numerical vector representing the importance of its words. These vectors can then be used as input for machine learning algorithms. TF-IDF is particularly effective in sentiment analysis because emotionally significant words such as “amazing”, “excellent”, “terrible”, or “boring” tend to receive higher weights and contribute strongly to sentiment classification.

7.2/ Sparse Matrix Representation

After TF-IDF vectorization, the dataset is transformed into a sparse numerical matrix.

In this matrix:

- Rows represent customer reviews
- Columns represent unique words or terms
- Values represent TF-IDF scores

Since each review contains only a limited subset of all possible words, most values in the matrix are equal to zero.

For example, a review discussing “action scenes” may not contain words related to “romance” or “comedy”. As a result, many columns remain empty for that review.

This type of representation is called a sparse matrix because the majority of values are zero.

Sparse matrices are highly efficient for handling large-scale textual datasets because they reduce memory usage and improve computational performance.

The use of sparse matrix representations is especially important in Big Data and NLP applications where datasets may contain thousands or millions of unique textual features.

7.3/ Saving TF-IDF Features Using Pickle

After generating the TF-IDF feature vectors, the transformed data and vectorization objects were saved using Python Pickle files.

Pickle is a Python serialization technique used to store complex Python objects in binary format. In this project, Pickle files were used to save the trained TF-IDF vectorizer and processed feature representations for future reuse.

Saving the TF-IDF data provided several important advantages:

- Avoiding repeated vectorization operations
- Reducing preprocessing time
- Improving workflow efficiency
- Facilitating model reuse and deployment
- Preserving the exact feature mapping used during training

Without saving the vectorizer, the entire TF-IDF transformation process would need to be repeated every time the project is executed.

By storing the vectorized data and preprocessing objects in Pickle format, the machine learning pipeline became more efficient and reproducible.

This approach is commonly used in machine learning projects because it allows preprocessing and feature engineering components to be reused consistently during testing, evaluation, or future deployment phases.

7.4/ Importance of Feature Engineering

Feature engineering plays a critical role in improving machine learning performance.

Well-designed features help machine learning models better understand textual patterns, contextual information, and sentiment-related expressions within customer reviews.

Without proper feature extraction, machine learning algorithms would struggle to interpret textual data effectively.

The combination of NLP preprocessing and TF-IDF vectorization provided several advantages:

- Improved representation of meaningful words
- Reduction of textual noise
- Better sentiment discrimination
- Improved computational efficiency
- Enhanced machine learning performance

Feature engineering also contributes to reducing dimensionality and simplifying the learning process for classification algorithms.

In this project, TF-IDF vectorization created a robust numerical representation of customer reviews, enabling the machine learning models to accurately classify positive, neutral, and negative sentiments.

8/ Data Preparation for Machine Learning

After feature extraction, the dataset was prepared for machine learning training and evaluation.

Proper data preparation is essential for building reliable predictive models and obtaining accurate evaluation results.

Several steps were performed to ensure that the dataset was correctly structured and consistent throughout the machine learning pipeline.

8.1/ Data Splitting

To evaluate model performance, the dataset was divided into:

- Training set: used to train the model
- Test set: used to evaluate performance

A standard split (80% training and 20% testing) was applied to ensure reliable evaluation.

8.2/ Training Dataset

The processed dataset was divided into training and testing subsets.

The training dataset contains the majority of customer reviews and is used during the learning phase of machine learning models.

During training, the algorithms analyze textual patterns and relationships between words and sentiment labels. The models learn how different terms and expressions contribute to positive, neutral, or negative sentiment predictions.

The quality and diversity of the training dataset directly influence model performance and prediction accuracy.

A sufficiently large training dataset allows the models to better generalize and recognize complex sentiment patterns in unseen reviews.

8.3/ Testing Dataset

The testing dataset contains reviews that are not used during the training process.

This dataset is essential for evaluating the real-world performance of the machine learning models. By testing the models on previously unseen data, it becomes possible to measure their ability to generalize beyond the training examples.

The testing phase helps evaluate several important aspects:

- Prediction accuracy
- Classification reliability
- Generalization capability
- Model robustness

Using separate testing data helps avoid overfitting, a situation where models memorize training examples instead of learning meaningful patterns.

Reliable testing procedures are essential for obtaining trustworthy evaluation results in machine learning projects.

9/ Big Data Processing with PySpark

PySpark was integrated into the project in order to demonstrate scalable Big Data processing capabilities on the Amazon Movies & TV Reviews dataset containing approximately 1.7 million customer reviews.

While Pandas provides efficient processing for moderate-sized datasets, it may encounter memory and performance limitations when handling very large-scale textual data. PySpark addresses these limitations by enabling distributed data processing across partitions, allowing large datasets to be processed more efficiently.

In this project, PySpark was used to reproduce several key performance indicators (KPIs) and aggregation operations initially implemented using traditional Python-based processing techniques. This approach demonstrates how distributed computing frameworks can support scalable analytics workflows in Big Data environments.

The integration of PySpark strengthens the Big Data aspect of the project by illustrating the transition from conventional in-memory processing toward scalable distributed data engineering techniques.

10/ Chapter Conclusion

This chapter presented the Big Data architecture and data engineering processes implemented throughout the project.

The chapter described the structure and characteristics of the Amazon Movies & TV Reviews dataset, the preprocessing pipeline, the Natural Language Processing techniques applied to customer reviews, and the feature engineering methods used to prepare the data for machine learning tasks.

Several technologies and tools were used, including Python, PySpark, NLTK, PyMongo, Scikit-learn, and Power BI. The integration of these technologies enabled efficient handling of large-scale textual data and facilitated the extraction of meaningful business insights.

The chapter also highlighted the importance of preprocessing, NLP operations, TF-IDF vectorization, and feature persistence using Pickle files in improving textual data quality and transforming unstructured reviews into numerical representations suitable for predictive modeling.

Finally, the dataset was prepared for machine learning through training and testing procedures designed to ensure reliable evaluation and prediction performance.

The next chapter presents the machine learning models implemented for sentiment classification, along with their training process, evaluation metrics, and comparative performance analysis.

Chapter 3:

Modeling & Prediction (Machine Learning)

1/ Introduction

Machine Learning plays a fundamental role in transforming customer review data into predictive and actionable insights. After completing the preprocessing and feature engineering phases described in the previous chapter, the dataset became suitable for sentiment classification tasks.

The objective of this chapter is to present the machine learning models used to classify customer reviews into positive, neutral, and negative sentiments based on textual content.

This chapter describes the machine learning workflow, the classification models implemented, the training and prediction process, and the evaluation techniques used to measure model performance.

Two supervised machine learning algorithms were implemented and compared in this project:

- Logistic Regression
- Linear Support Vector Classifier (Linear SVC)

The models were trained using TF-IDF feature vectors extracted from customer reviews. Their predictive performance was evaluated using multiple evaluation metrics in order to identify the most effective sentiment classification model.

2/ Machine Learning Workflow

The machine learning workflow implemented in this project follows a structured pipeline designed to transform raw customer reviews into sentiment predictions.

The workflow consists of several sequential stages that ensure proper preparation, transformation, training, and evaluation of the data.

The overall machine learning pipeline can be summarized as follows:

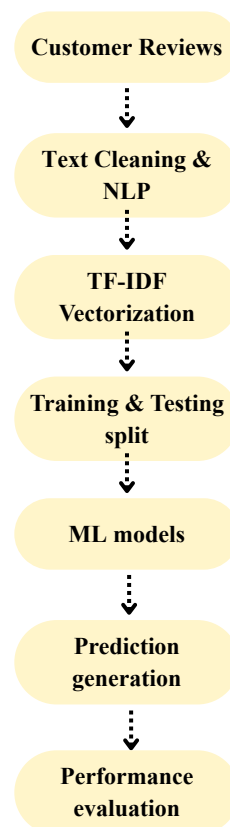


Figure 3.1: Machine Learning Workflow Pipeline

This pipeline allows the system to process unstructured textual reviews and automatically predict customer sentiment.

2.1 Input Features and Labels

The machine learning models require numerical input features and target labels.

Input Features

The input features used in this project were generated using TF-IDF vectorization applied to the cleaned customer reviews.

Each review was transformed into a numerical vector representing the importance of words within the dataset.

These vectors allowed the machine learning models to analyze textual patterns and sentiment-related expressions.

Target Labels

The target variable used for classification was the sentiment column generated during preprocessing.

The sentiment labels were defined as follows:

- Positive → ratings greater than or equal to 4
- Neutral → rating equal to 3
- Negative → ratings less than or equal to 2

The objective of the models was therefore to predict the correct sentiment category based on review text.

2.2 Machine Learning Pipeline

The machine learning pipeline implemented in this project consists of multiple integrated stages.

Step 1: Text Cleaning and NLP

Customer reviews first undergo preprocessing and NLP operations, including:

- Lowercasing
- Stopword removal
- Negation handling
- Removal of punctuation and special characters

This stage improves text quality and reduces noise.

Step 2: TF-IDF Vectorization

The cleaned reviews are converted into numerical feature vectors using TF-IDF vectorization.

This transformation allows machine learning algorithms to process textual information mathematically.

Step 3: Dataset Splitting

The vectorized dataset is divided into training and testing subsets.

The training subset is used to train the models, while the testing subset is reserved for evaluation.

Step 4: Model Training

Machine learning algorithms learn relationships between textual patterns and sentiment labels.

Step 5: Prediction and Evaluation

The trained models generate predictions on unseen testing data, and their performance is evaluated using multiple metrics.

2.3 Sentiment Classification Objective

The primary objective of the machine learning system is to automatically classify customer reviews into sentiment categories.

Sentiment classification provides several business advantages:

- Understanding customer satisfaction
- Detecting negative customer experiences
- Identifying trends in customer opinions
- Supporting decision-making processes
- Improving products and services

Automating sentiment analysis enables organizations to process large volumes of customer feedback more efficiently than manual analysis.

This project demonstrates how machine learning techniques can transform unstructured customer opinions into valuable business intelligence insights.

3/ Model Selection

Two supervised machine learning algorithms were selected and implemented for sentiment classification.

The selected models are widely used in Natural Language Processing and text classification tasks because of their efficiency and strong performance with high-dimensional textual data.

3.1/ Logistic Regression

Logistic Regression is a supervised machine learning algorithm commonly used for classification problems.

Despite its name, Logistic Regression is primarily a classification model rather than a regression technique.

The model estimates the probability that a review belongs to a specific sentiment category.

Logistic Regression is particularly effective for text classification because:

- It performs well with sparse textual features
- It handles high-dimensional data efficiently
- It provides fast training and prediction times
- It works effectively with TF-IDF representations

The model attempts to identify linear relationships between textual features and sentiment categories.

In this project, Logistic Regression was trained using TF-IDF feature vectors extracted from customer reviews.

3.2 Linear Support Vector Classifier (Linear SVC)

Linear Support Vector Classifier (Linear SVC) is another supervised machine learning algorithm frequently used for text classification and sentiment analysis tasks.

Linear SVC attempts to find an optimal decision boundary that separates sentiment classes while maximizing the distance between categories.

The model offers several advantages:

- Strong performance on high-dimensional datasets
- Efficient handling of sparse matrices
- Good generalization capability
- High effectiveness for textual classification problems

Linear SVC is widely recognized as one of the most effective algorithms for Natural Language Processing applications.

The model was trained using the same TF-IDF feature vectors used for Logistic Regression.

3.3 Why These Models Were Selected

The choice of Logistic Regression and Linear SVC was motivated by several factors.

Both models:

- Perform efficiently on textual datasets
- Handle sparse TF-IDF matrices effectively
- Provide strong classification accuracy
- Require relatively low computational resources
- Are widely used in sentiment analysis research

Additionally, these models are particularly suitable for large-scale customer review datasets because they maintain strong predictive performance while remaining computationally efficient.

The use of two different models also allowed comparative analysis in order to identify the most effective sentiment classification approach.

4/ Model Training

After selecting the machine learning algorithms, the models were trained using the prepared TF-IDF feature vectors and sentiment labels.

The training process enables the algorithms to learn relationships between textual patterns and sentiment categories.

4.1 Training the Logistic Regression Model

The Logistic Regression model was trained using the training subset of the dataset.

During training, the model analyzed the relationships between TF-IDF features and customer sentiment labels.

The objective of the model was to minimize classification errors and improve prediction accuracy.

Once training was completed, the model became capable of predicting the sentiment category of unseen customer reviews.

4.2/ Training the Linear SVC Model

The Linear SVC model was also trained using the same training dataset.

The model attempted to identify optimal boundaries separating positive, neutral, and negative sentiment classes.

Linear SVC is particularly effective for sentiment analysis because it performs well with sparse and high-dimensional textual data.

The model learned discriminative textual patterns associated with different sentiment categories.

4.3/ Generating Predictions

After training, both models generated predictions using the testing dataset.

The predicted sentiments were compared with the actual sentiment labels in order to evaluate model performance.

Prediction generation represents the final stage of the classification process, where machine learning models apply learned patterns to previously unseen customer reviews.

5/ Model Evaluation

Model evaluation is essential for measuring the effectiveness and reliability of machine learning systems.

Several evaluation metrics were used in this project to compare model performance and identify the best sentiment classification approach.

5.1/ Accuracy Score

Accuracy measures the percentage of correctly classified reviews.

It is calculated as the ratio between correct predictions and total predictions.

Higher accuracy values indicate better classification performance.

Accuracy provides an overall measure of model effectiveness but should be interpreted alongside additional evaluation metrics.

Logistic Regression Accuracy : 81.89%

Linear SVC Accuracy : 81.82%

=> Both models have a close accuracy

5.2/ Classification Report

The classification report provides detailed evaluation metrics for each sentiment category.

The report includes:

- Precision
- Recall
- F1-score
- Support

These metrics provide deeper insights into model performance and classification quality.

The classification report is particularly useful for identifying strengths and weaknesses across different sentiment categories.

5.3/ Confusion Matrix

The confusion matrix is a visual evaluation tool used to analyze prediction performance.

It compares actual sentiment labels with predicted sentiment labels.

The confusion matrix helps identify:

- Correct classifications
- Misclassification patterns
- Model weaknesses
- Sentiment confusion between categories

This visualization provides a detailed understanding of classification behavior.

6/ Best Model Selection

After evaluating both machine learning models, the best-performing model was selected based on overall classification performance.

The selection process considered:

- Accuracy
- Stability
- Generalization capability
- Performance across sentiment categories

Although Logistic Regression achieved the highest classification accuracy with approximately 81.89%, the performance difference compared to Linear SVC, which achieved approximately 81.82%, remains relatively small.

This result indicates that both models are highly effective for sentiment classification tasks involving sparse textual data represented using TF-IDF features.

The strong performance of both algorithms can be explained by their suitability for high-dimensional Natural Language Processing datasets. Since TF-IDF vectorization generates sparse feature matrices containing thousands of textual attributes, both Logistic Regression and Linear SVC are particularly well adapted to this type of data representation.

Despite the slightly superior performance of Logistic Regression, the experimental results demonstrate that both models provide reliable and efficient sentiment classification capabilities for large-scale customer review analysis.

7/ Challenges and Limitations

Despite the strong performance of the models, several challenges and limitations were encountered during the project.

Contextual Complexity

Human language is highly complex and context-dependent. Certain reviews may contain sarcasm, irony, or mixed sentiments that are difficult for machine learning models to interpret accurately.

Sentiment Ambiguity

Some customer reviews contain both positive and negative opinions simultaneously, making classification more difficult.

For example:

"The movie had good visuals but a weak storyline."

Such reviews may contain mixed emotional signals.

Rating-Based Sentiment Labels

The sentiment labels used in this project were generated from numerical ratings rather than manually annotated sentiment labels.

As a result, certain reviews may contain inconsistencies between textual sentiment and assigned labels.

NLP Limitations

Traditional NLP preprocessing techniques may not fully capture contextual meaning, emotional nuance, or linguistic complexity.

These limitations highlight the challenges involved in large-scale sentiment analysis tasks.

8/ Chapter Conclusion

This chapter presented the machine learning models and predictive techniques implemented for sentiment classification.

The chapter described the machine learning workflow, the TF-IDF-based feature representation, the supervised classification models used, and the evaluation procedures applied to measure predictive performance.

Two machine learning models, Logistic Regression and Linear SVC, were trained and evaluated using customer review data from the Amazon Movies & TV dataset.

The results demonstrated the effectiveness of machine learning techniques for automatically analyzing customer sentiment and extracting valuable insights from large-scale textual data.

The next chapter focuses on the visualization, business intelligence, and analytical dashboards developed using Power BI to transform predictive results into actionable business insights.

Chapter 4:

Visualization & Business Analysis

1/ Introduction

This chapter presents the visualization and interpretation of the results obtained from the data analysis and machine learning phases. The objective is to transform processed data and model outputs into meaningful insights that can support decision-making.

To achieve this, an interactive dashboard was developed using Microsoft Power BI. The dashboard allows users to explore customer reviews, analyze sentiment, evaluate product performance, and observe trends over time.

2/ Dashboard Overview

The dashboard is designed to provide a clear and structured view of the data through multiple pages, each focusing on a specific aspect of the analysis. It enables users to interact with the data using filters and visual elements, making it easier to extract insights.

The dashboard is divided into four main pages:

- Overview
- Customer Review Insights
- Product Performance
- Review Trends

Each page contains a combination of Key Performance Indicators (KPIs) and visualizations tailored to highlight specific patterns and metrics.

3/ Dashboard Structure

3.1/ Overview Page

The Overview page provides a high-level summary of the dataset.

Key indicators include:

- **Total Number of Reviews:** represents the total volume of customer feedback
- **Average Rating:** provides an overall measure of customer satisfaction

The overview highlights a general increase in review volume over time, accompanied by a high proportion of positive sentiment and a stable average rating, indicating overall customer satisfaction.

This page gives users a quick understanding of the scale and general sentiment of the dataset.

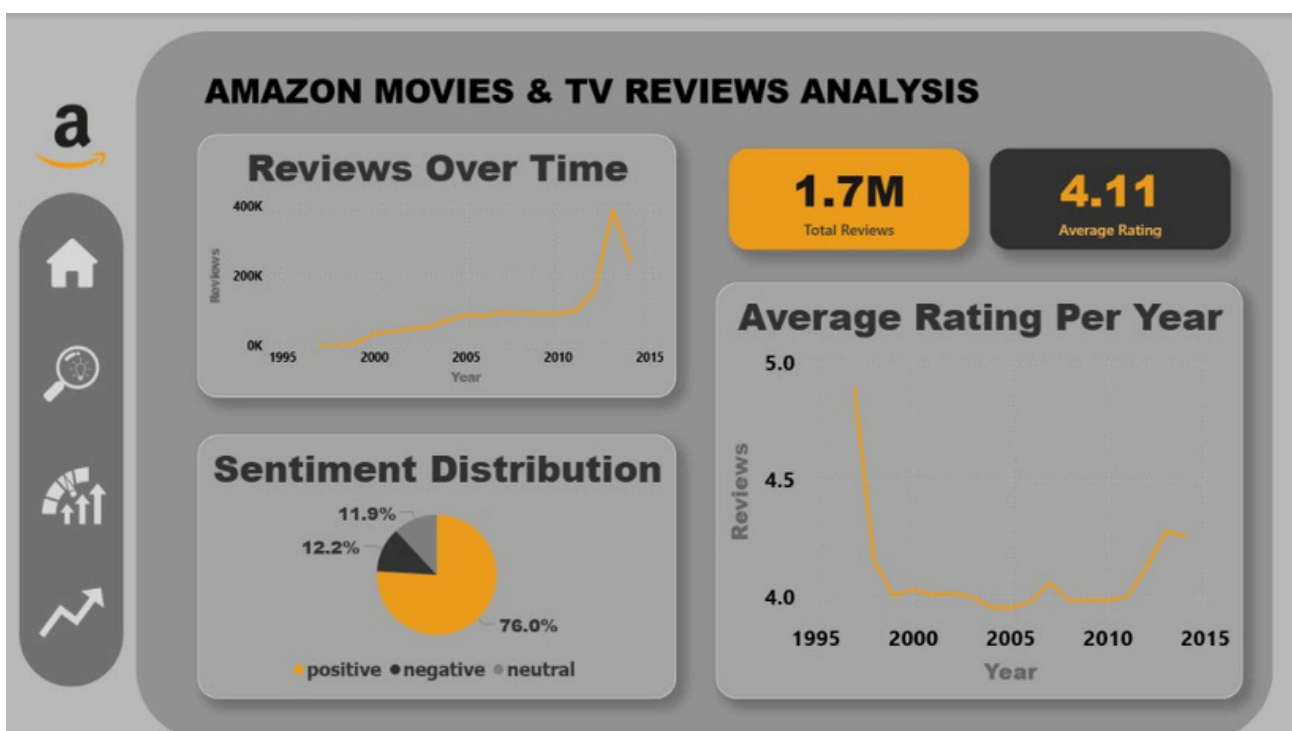


Figure 4.1 - Overview Dashboard Page

3.2/ Customer Review Insights Page

This page focuses on analyzing the content of customer reviews and extracting insights related to sentiment and textual characteristics.

Key KPIs include:

Positive Rate (%): percentage of reviews classified as positive

Negative Rate (%): percentage of reviews classified as negative

Average Review Length: average number of words per review

The results show that approximately 76% of reviews are positive, while only about 12% are negative. This reflects a strong overall customer satisfaction level. Additionally, the relatively high average review length suggests that users are actively engaged and tend to provide detailed feedback.

Key visualizations include:

Most Frequent Words Analysis

A bar chart visualization highlights the most commonly used words in customer reviews.

The most frequent terms include words such as “like”, “good”, “great”, and “well”, which are predominantly positive in nature. This confirms the sentiment distribution observed in the dataset. Furthermore, the presence of words such as “story” and “time” indicates that customers often focus on content quality and viewing experience when sharing their opinions.

Rating Distribution

The rating distribution chart illustrates how customer ratings are distributed across different values. The analysis shows a strong concentration of high ratings, particularly at 4 and 5 stars, with 5-star reviews being the most dominant. In contrast, low ratings (1 and 2 stars) are relatively rare. This distribution further confirms the overall positive perception of the products.

Review Length Distribution by Sentiment

This visualization examines how the length of reviews varies across different sentiment categories. The results indicate that longer reviews are more common, especially among positive sentiments. This suggests that satisfied customers tend to provide more detailed and expressive feedback. Shorter reviews, on the other hand, are less frequent and may reflect lower engagement or simpler opinions.

Overall Insight:

Overall, the Customer Review Insights page highlights a highly positive customer perception, supported by both sentiment and rating distributions. The frequent use of positive language and the prevalence of longer, detailed reviews indicate strong user engagement and satisfaction with the products.

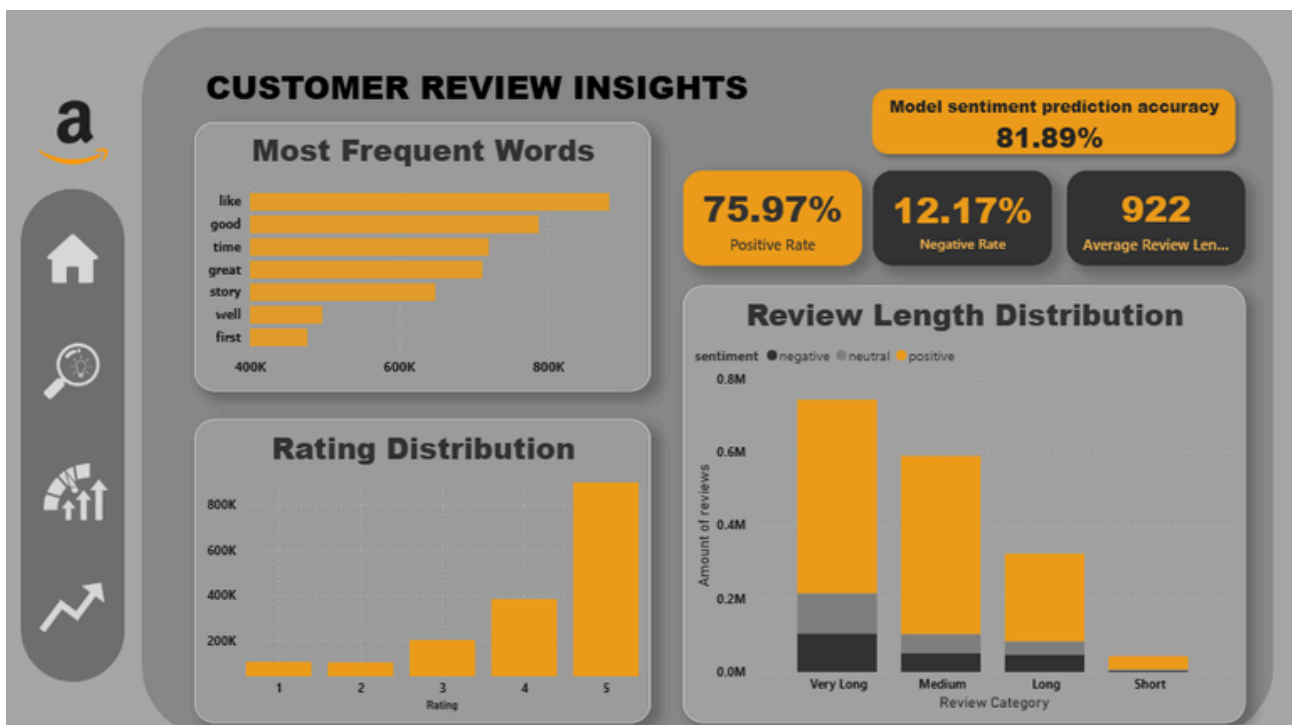


Figure 4.2 - Customer Review Insights Page

3.3/ Product Performance Page

The Product Performance page focuses on evaluating individual products based on customer reviews.

Key visualizations include:

Top Rated Products

The best rated products all achieve ratings close to 5, indicating exceptionally high customer satisfaction. These products consistently meet or exceed user expectations, suggesting strong quality, content value, and positive user experience. However, it is important to note that high ratings do not necessarily imply high popularity, as some of these products may have fewer reviews.

Worst Rated Products

The worst rated products have significantly lower average ratings, in some cases close to 1.0, indicating strong customer dissatisfaction. These products may suffer from poor quality, misleading expectations, or negative user experiences. Identifying such products is critical for businesses, as they highlight areas requiring improvement or potential removal from the platform.

Top Reviewed Products

The top reviewed products have the highest number of customer reviews, indicating strong popularity and high user engagement. These products attract significant attention, which may be due to brand recognition, content popularity, or marketing efforts. However, a high number of reviews does not necessarily guarantee high satisfaction, as popularity and quality are not always correlated.

Overall, the analysis of product performance highlights a clear distinction between popularity and satisfaction. While some products receive a high volume of reviews, others achieve higher ratings with fewer reviews. This suggests that businesses should consider both review volume and average rating when evaluating product success and making strategic decisions.

I wanted to add a visual in this page that will show which categories (ex:drama,action,...) get the most reviews,however the dataset didn't include categories.



Figure 4.3 - Product Performance Analysis Page

3.4/ Review Trends Page

This page analyzes how customer reviews evolve over time.

Key KPIs include:

Average Monthly Review Volume (2014)

The average monthly review volume in 2014 is approximately 34.89K reviews, reflecting a consistently high level of user engagement throughout the year despite the observed decline after the initial peak.

Peak Review Month (2014)

January is identified as the peak review month in 2014, with approximately 49.28K reviews. This indicates a period of maximum user activity, which may be associated with post-holiday engagement or increased product consumption during that time.

Visualizations include:

Monthly Review Activity (2014)

The monthly review activity in 2014 shows a clear declining trend, starting with a peak in January and gradually decreasing towards July. This suggests that customer engagement was highest at the beginning of the year, possibly due to seasonal factors such as holiday purchases, and declined over time as user activity stabilized.

Average Rating Over Time

The average rating over time shows an initial decline from higher values, followed by a long period of stability around a rating of 4.0. In later years, a slight upward trend is observed, suggesting a gradual improvement in customer satisfaction and product quality.

Sentiment Over Time

The sentiment over time analysis shows that positive reviews consistently dominate across all periods, with a significant increase in volume in later years. Negative and neutral sentiments remain relatively low in comparison. This indicates that the growth in review volume is largely driven by positive customer experiences, reinforcing the overall satisfaction trend observed in the dataset. This page provides insights into temporal trends and customer engagement patterns.

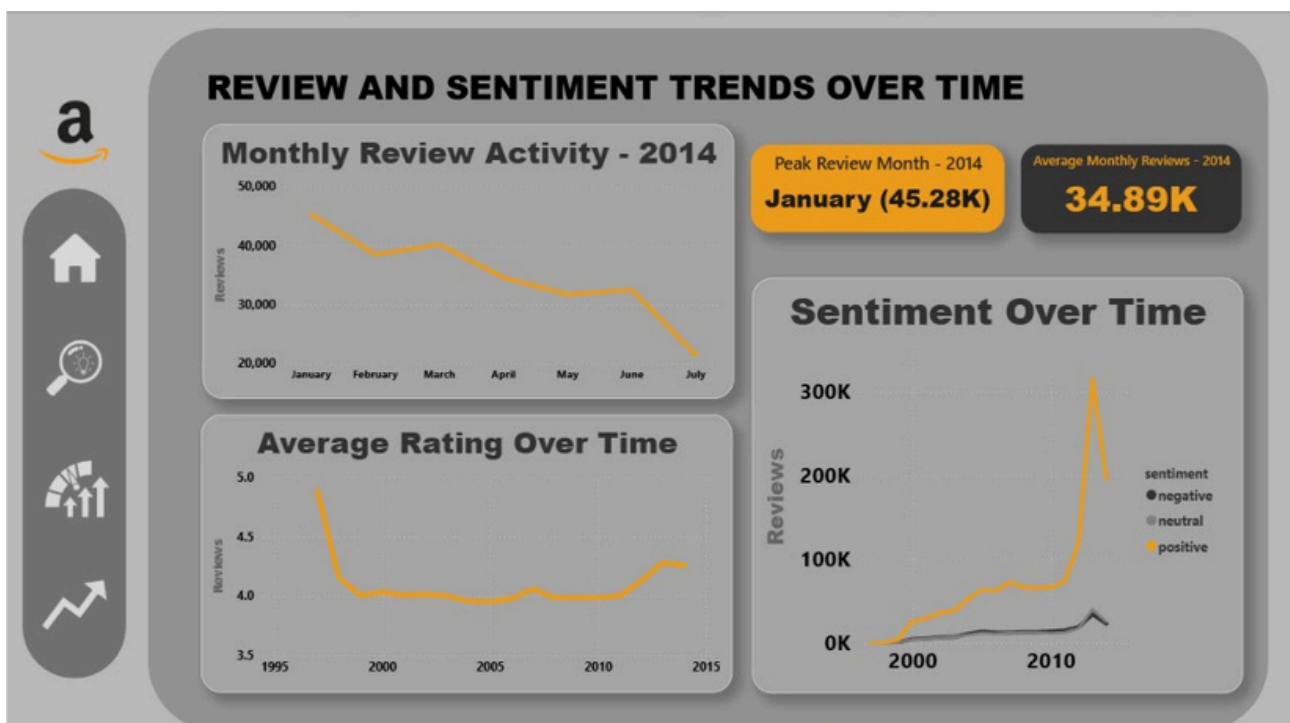


Figure 4.4 - Review and Sentiment Trends Over Time

4/ Business Insights

The analysis of the dataset reveals several important insights:

- A majority of reviews are positive, indicating generally high customer satisfaction
- Products with a high number of reviews tend to attract more visibility and engagement
- Certain periods show increased review activity, suggesting seasonal or promotional effects
- Review length varies depending on sentiment, with more detailed reviews often expressing stronger opinions

These insights can help businesses better understand customer behavior and improve their products and services.

5/ Decision-Making Impact

The results of this project can support decision-making in several ways:

- Product Improvement: identifying poorly rated products helps prioritize improvements
- Customer Understanding: sentiment analysis reveals customer expectations and satisfaction levels
- Marketing Strategies: analyzing review trends can help optimize promotional campaigns
- Performance Monitoring: dashboards allow continuous tracking of customer feedback

By leveraging these insights, companies can make more informed and data-driven decisions

6/ Limitations

Despite the effectiveness of the dashboard, certain limitations exist:

The dataset contains incomplete or missing product titles in some cases

Sentiment labels are derived from ratings and may not always reflect actual opinions

The analysis is limited to historical data and does not include real-time updates

Some linguistic nuances in reviews may not be fully captured

These limitations should be considered when interpreting the results.

7/ Future Improvements

Several enhancements can be made to improve the system:

- Integration of real-time data sources
- Use of more advanced visualization techniques
- Implementation of more sophisticated sentiment analysis models
- Expansion to other product categories

These improvements would increase the scalability and effectiveness of the solution.

8/ Chapter Conclusion

This chapter presented the visualization and interpretation of the results through an interactive dashboard. It demonstrated how data analysis and machine learning outputs can be transformed into actionable insights.

The dashboard provides a comprehensive view of customer reviews, enabling better understanding of customer sentiment, product performance, and review trends. Overall, this project highlights the value of combining Big Data technologies, machine learning, and data visualization to support informed decision-making.

General Conclusion

This project presented the development of a Big Data analytics and sentiment analysis system based on the Amazon Movies & TV Reviews dataset.

The main objective was to analyze large-scale customer reviews, automatically classify customer sentiment, and generate business-oriented insights through interactive visualizations.

Several stages were implemented throughout the project, beginning with data collection, preprocessing, and cleaning operations.

The dataset was enriched by integrating metadata information and preparing textual reviews for analysis using Natural Language Processing techniques.

Important preprocessing tasks such as stopwords removal, negation handling, text normalization, and sentiment label generation significantly improved data quality and prepared the dataset for machine learning applications.

Feature engineering was performed using TF-IDF vectorization in order to transform unstructured textual reviews into numerical representations suitable for predictive modeling. Two machine learning models, Logistic Regression and Linear SVC, were trained and evaluated for sentiment classification tasks.

The results demonstrated the effectiveness of machine learning techniques in automatically identifying customer sentiment from textual reviews.

In addition, the project integrated Big Data technologies such as PySpark and MongoDB to demonstrate scalable data processing capabilities within a distributed data environment.

Interactive dashboards developed using Power BI enabled the visualization of review trends, sentiment distributions, customer behavior, and product performance indicators. These dashboards transformed analytical results into understandable and actionable business insights.

Despite the positive results obtained, several challenges were encountered, particularly regarding contextual sentiment interpretation, sarcasm detection, and inconsistencies between textual sentiment and rating-based labels. These limitations highlight the complexity of Natural Language Processing and sentiment analysis tasks.

Future improvements could include the integration of deep learning models, advanced NLP techniques, real-time data processing systems, and more sophisticated contextual language understanding approaches.

Overall, this project demonstrates how Big Data technologies, Natural Language Processing, Machine Learning, and Business Intelligence tools can be combined to extract valuable knowledge from large-scale customer review data and support data-driven decision-making processes.

Appendix

Appendix A : NLP Preprocessing Code

```
34 # NLP TOOLS
35
36 stop_words = set(stopwords.words("english")) # set the language of the stopwords we wanna remove to english, makes the search faster
37 # Remove important words from stopwords list
38 important_words = {"not", "no", "nor", "but"}
39 stop_words = stop_words - important_words
40
41 lemmatizer = WordNetLemmatizer() # convert for example running → run , cars → car , better → good
42
43 # CLEANING FUNCTION (this function cleans 1 review at a time)
44
45 def preprocess_text(text):
46
47     if pd.isna(text): # if review is empty ,return empty string (avoid errors)
48         return ""
49
50     # Lowercase
51     text = text.lower()
52
53     # Remove punctuation & numbers
54     text = re.sub(r'[^a-zA-Z\s]', '', text)
55
56     # Tokenization
57     tokens = word_tokenize(text) # split sentence into words (example : "this product is amazing" → ["this", "product", "is", "amazing"])
58
59     # Remove stopwords
60     tokens = [word for word in tokens if word not in stop_words] # ["this", "product", "is", "amazing"] → ["product", "amazing"]
61
62     # Lemmatization
63     tokens = [lemmatizer.lemmatize(word) for word in tokens] # ["products", "running"] → ["product", "run"]
64
65     return " ".join(tokens) # join tokens back to a sentence (["product", "amazing"] → "product amazing")
66
67
```

Appendix B : TF-IDF & Feature Engineering

```
phase3_TF-IDF.py X
src > phase3_TF-IDF.py > ...
28
29 # Sample dataset for faster TF-IDF
30 print("Sampling 200,000 reviews...")
31 df = df.sample(200000, random_state=42)
32
33 print("Sampled dataset shape:", df.shape)
34
35 # Text column
36 X_text = df["clean_text"]
37
38 # Target column
39 y = df["sentiment"]
40
41 print("Creating TF-IDF vectors...")
42
43 vectorizer = TfidfVectorizer(
44     max_features=5000,
45     stop_words="english",
46     min_df=5
47 )
48
49 X = vectorizer.fit_transform(X_text)
50 print("TF-IDF completed")
51 print("Feature matrix shape:", X.shape)
52
53 # Save Processed Data
54 pickle.dump((X, y, vectorizer), open("data/processed_data.pkl", "wb"))
55 print("Phase 3 data saved successfully")
```

Appendix C : Machine Learning Models

```
phase4_modelML.py X
src > phase4_modelML.py > ...
6 # Load Phase 3 Data
7 X, y, vectorizer = pickle.load(open("data/processed_data.pkl", "rb"))
8
9 print("Data loaded successfully")
10
11 # Train / Test Split
12 X_train, X_test, y_train, y_test = train_test_split(
13     X, y, test_size=0.2, random_state=42
14 )
15
16 print("Data split completed")
17
18 # Train Models
19 # Logistic Regression
20 lr_model = LogisticRegression(max_iter=1000)
21 lr_model.fit(X_train, y_train)
22 lr_preds = lr_model.predict(X_test)
23 lr_accuracy = accuracy_score(y_test, lr_preds)
24 print("Logistic Regression Accuracy:", lr_accuracy)
25
26 # Linear SVC
27 svc_model = LinearSVC()
28 svc_model.fit(X_train, y_train)
29 svc_preds = svc_model.predict(X_test)
30 svc_accuracy = accuracy_score(y_test, svc_preds)
31 print("LinearSVC Accuracy:", svc_accuracy)
32
33 # Show Report of each model
34 print("\nLogistic Regression Report:")
35 print(classification_report(y_test, lr_preds))
36
37 print("\nLinearSVC Report:")
38 print(classification_report(y_test, svc_preds))
39
40 # Choose the best model (best accuracy)
41 if svc_accuracy > lr_accuracy:
42     best_model = svc_model
43     model_name = "LinearSVC"
44 else:
45     best_model = lr_model
46     model_name = "LogisticRegression"
47
48 print("Best model selected:", model_name)
49
50 # Save best model
51 pickle.dump(best_model, open("models/sentiment_model.pkl", "wb"))
52 pickle.dump(vectorizer, open("models/vectorizer.pkl", "wb"))
```

Appendix D : Project Structure

